David Lin

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https://tealeave.github.io/da-wei-lin-data/

SUMMARY

Performance-focused Data Scientist with a Ph.D. in Tumor Metabolism and over eight years of experience architecting and deploying scalable pipelines for genomics, clinical research, and multi-omics datasets. Proficient in cutting-edge neural network designs (including convolutional and transformer architectures), deep learning methods, and data analytics. A trusted collaborator with wet-lab and ML scientists, driving quality control, workflow optimization, biomarker discovery, and actionable data insights. Passionately engaged with the latest advances in multi-omic modeling, diffusion- and language-model-driven applications.

CORE COMPETENCIES

- **AI / ML Frameworks & Algorithms:** Python ecosystem (NumPy, pandas, scikit-learn, **TensorFlow/Keras, PyTorch**); extensive experience with **MLOps** pipelines and graph/vector databases (**Neo4j, Redis**) integrated with **LLM**s. Currently building a **multi-agent clinical NLP system** for medical-necessity decisioning using the MIMIC-IV **EHR** dataset (under PhysioNet credentialing).

- **Data Domains & Bioinformatics:** Designed and validated end-to-end **bioinformatics** pipelines (**germline, somatic, RNA-seq, proteomics, metabolomics**); performed feature selection; drafted statistical analysis plans and coordinated sample workflows; ensured quality management and regulatory compliance.

- **DevOps, Cloud & HPC: Docker** microservices and networking; **GitHub** code review and CI/CD; pipeline versioning, validation, and performance monitoring; **AWS** (S3, EC2); virtual machines; databases (**PostgreSQL**, **Neo4j**—graph algorithms, **MongoDB**); **SLURM**-based **HPC** orchestration; **Nextflow** pipeline orchestration; containerized cluster deployment.

- **Programming & Data Engineering:** Advanced skills in **Python** and **R**; **SQL** and **PySpark** for large-scale ELT and high-dimensional clinical data processing.

- **Wet-lab Experience:** Proficient with molecular biology methods, vector design, and HTS reporter systems in yeast and mammalian cells; familiar with **NGS** library prep.

- **Deep Learning Models:** Expert in deep learning architectures including **CNNs**, **Vision Transformers (ViTs)**, and **multimodal** transformer models (early or late fusion); **XGBoost for tabular data**; proficient in model fine-tuning, ensemble methods, **experiment tracking with MLflow**, and generative AI applications for image or sentiment analysis.

- **Collaboration & Agile Delivery:** Agile methodologies with **Jira**; cross-functional teamwork with wet-lab scientists, ML scientists, and clinicians; managing stakeholder expectations.

EXPERIENCE

Data Scientist III

Sapient Bioanalytics **Oct 2023 - Feb 2025**

- **Multi-Omics Pipelines**: Architected a three-stage PySpark pipeline for a 50k-sample metabolomics study (40k+ features/sample)—cleaning, feature engineering, and regression—supporting the Gates Foundation's MOMI project across LMIC sites.
- **Metabolic Risk Scoring**: Developed an XGBoost-driven scoring system with Optuna-optimized hyperparameters; deployed on Databricks to process population-scale datasets.

- **Capacity Building Leadership**: Designed and delivered training workshops in developing countries on best practices for omics analysis, distributed computing, EDA, and ML/AI methodology, empowering global researchers with advanced data capabilities.
- **Cross-Functional Collaboration:** Coordinated with mass spectrometry, computational chemistry, and clinical scientists to ensure data integrity, reproducibility, and alignment with project milestones.

Bioinformatic Scientist II - III

Ambry Genetics **Dec 2019 - Oct 2023**

- **Panel Development**: Orchestrated the development of high-volume NGS panels (CancerNext, RNA, WES, and somatic/oncology panels) and subsequently conducted the validation. Ensured ≥99% sensitivity and specificity for SNVs, indels and CNVs under regulatory guidelines. Served as **CNV algorithm SME** during FDA submission.
- **QC & Statistical Thresholding**: Established robust statistical cutoffs for allele-frequency and coverage metrics, reducing false positives by 30% during chemistry transitions.
- **Pipeline Performance Monitoring:** Developed real-time dashboards to track and visualize variant-calling pipeline performance.
- **Machine Learning Applications**: Implemented GradientBoostingRegressor models to predict CNV counts and prioritize QC investigations, improving workflow efficiency.

Postdoctoral Researcher

UC Irvine **Jan 2014 - Dec 2019**

- **CRISPR & Proteomics**: Developed CRISPR tagging/knockout tools and purified tagged PP2A complexes for SILAC-LC-MS; mapped protein–protein interactions under metabolic stress.
- **HTS Bio-Screening:** Developed high-throughput screening platform in mammalian cells for p53-reactivating compounds, resulting in a publication in *Nature Communications*.
- **RNA-seq and splicing analysis pipeline:** Designed **Nextflow** pipelines for QC, preprocessing, differential expression, and splicing analysis, identifying methionine-stress targets for therapeutics.

EDUCATION

University of California, Irvine **Sep 2007 – Jan 2014**
Ph.D. Biological Sciences
University of California, Berkeley
M.S. Information and Data Science **In progress**

AWARDS & PUBLICATIONS

Molecular Metabolism | Nutrient control of splice site selection contributes to methionine addiction of cancer **2025**

Journal of Lipid Research | Lipid remodeling in response to methionine stress in MDA-MBA-468 triple-negative breast cancer cells **2021**

Journal of Biological Chemistry | Microhomology-based CRISPR tagging tools for protein tracking, purification, and depletion **2019**

Methods Mol. Biol. | Isolation and characterization of methionine-independent clones from methionine-dependent cancer cells **2019**

U.S. Patent | ChemBridge Small Molecules that could enhance p53 activity **2015**

Journal of Cell Science | SAM limitation induces p38 mitogen-activated protein kinase and triggers cell cycle arrest in G1 **2014**

Nature Communications | Computational identification of a transiently open L1/S3 pocket for reactivation of mutant p53 **2013**